Fundamentals of Software Engineering: A Comprehensive Guide

Introduction to Software Engineering Principles

Software engineering is the systematic application of engineering principles to the development of software. Unlike simple programming, software engineering involves the entire lifecycle of a product, from initial conception to maintenance and retirement. The primary goal is to produce high-quality software that is reliable, efficient, and maintainable within budget and time constraints.

Core Engineering Principles

The following table summarizes the fundamental principles that guide robust software development:

Principle	Description	Application in Practice
Separation of Concerns	Dividing a computer program into distinct sections such that each section addresses a separate concern.	Layered architecture (e.g., separating UI from business logic).
Modularity	The degree to which a system's components may be separated and recombined.	Using microservices or libraries to build complex systems.
Abstraction	Reducing information and detail to focus on concepts relevant to a particular purpose.	Using APIs or interfaces to interact with complex subsystems.
Anticipation of Change	Designing software with the expectation that requirements and environments will evolve.	Implementing design patterns like Strategy or Observer.
Generality	Designing a solution to be applicable to a range of problems rather than a single specific instance.	Developing generic classes or reusable utility functions.
Incremental Development	Building software in small, manageable parts, with each part adding new functionality.	Agile sprints and continuous delivery practices.
Consistency	Ensuring that the software follows a uniform style, structure, and set of rules.	Adhering to coding standards and architectural patterns.

The SOLID Principles

For object-oriented design, the **SOLID** acronym represents five key principles that enhance maintainability and scalability:

1. **Single Responsibility Principle (SRP):** A class should have only one reason to change, meaning it should perform only one primary task.
2. **Open/Closed Principle (OCP):** Software entities should be open for extension but closed for modification.
3. **Liskov Substitution Principle (LSP):** Objects of a superclass should be replaceable with objects of its subclasses without affecting the correctness of the

program.

4. **Interface Segregation Principle (ISP):** Clients should not be forced to depend on interfaces they do not use.
5. **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules; both should depend on abstractions.

Software Processes and State-of-the-Art Models

A software process is a structured set of activities required to develop a software system. These activities typically include specification, design and implementation, validation, and evolution. Selecting the right process model is critical to project success.

Traditional vs. Modern Models

Model Category	Examples	Key Characteristics
Plan-Driven (Traditional)	Waterfall, V-Model	Linear, sequential phases; heavy documentation; difficult to change requirements late.
Iterative/Evolutionary	Spiral Model	Risk-driven; multiple iterations; incorporates prototyping and risk analysis.
Agile Methodologies	Scrum, Kanban, XP	Adaptive planning; early delivery; continuous improvement; rapid response to change.
State-of-the-Art	DevOps, DevSecOps	Integration of development and operations; emphasis on automation, CI/CD, and security.

> ***Note on DevOps:*** *Modern software engineering increasingly relies on DevOps practices. This approach breaks down silos between development and operations teams, utilizing automated pipelines for testing and deployment to increase the velocity of software delivery.*

Data Communication, Computer Networks, and Security

Understanding how data moves across systems is essential for modern software engineers, as most applications are now distributed or cloud-based.

Fundamentals of Data Communication

Data communication involves the exchange of data between two nodes via some form of transmission medium. A standard communication system consists of five components: the **message** (data), the **sender**, the **receiver**, the **medium** (physical path), and the **protocol** (rules).

Networking Models and Architecture

The **OSI (Open Systems Interconnection) Model** and the **TCP/IP Model** are the primary frameworks for understanding network interactions.

Layer	OSI Model Layer	TCP/IP Model Layer	Primary Function
7	Application	Application	Network process to application.
6	Presentation	Application	Data representation and encryption.
5	Session	Application	Interhost communication.
4	Transport	Transport	End-to-end connections and reliability (TCP/UDP).
3	Network	Internet	Path determination and logical addressing (IP).
2	Data Link	Network Access	Physical addressing (MAC) and error detection.
1	Physical	Network Access	Media, signal, and binary transmission.

Network Security and the CIA Triad

Security must be integrated into every phase of the software engineering process. The core objectives of security are defined by the **CIA Triad**:

- **Confidentiality:** Ensuring that data is accessible only to those authorized to have access.
- **Integrity:** Maintaining the accuracy and consistency of data over its entire lifecycle.
- **Availability:** Ensuring that information and resources are available to users when needed.

Installation, Configuration, and Engineering Tools

Software engineers must be proficient in setting up their environments and utilizing tools that automate and manage the development process.

System Software and Configuration

Installation of system software involves more than just running an installer. It requires configuring environment variables (like `PATH`), managing dependencies, and ensuring compatibility between the operating system and development libraries. In a network context, this includes configuring IP addresses, subnet masks, and default gateways to enable communication within a Local Area Network (LAN).

Essential Software Engineering Tools

Tool Category	Common Examples	Purpose
Version Control	Git, SVN	Tracking changes in source code and enabling collaboration.
IDE	VS Code, IntelliJ, Eclipse	Providing a comprehensive environment for writing and debugging code.
Build Automation	Maven, Gradle, Jenkins	Automating the compilation, testing, and packaging of software.
Project Management	Jira, Trello, Azure DevOps	Managing tasks, backlogs, and team workflows.
Testing Frameworks	JUnit, Selenium, PyTest	Ensuring code quality through automated unit and integration tests.

Professionalism, Ethics, and Norms

Software engineering is a profession that carries significant social responsibility. The **ACM/IEEE Code of Ethics** provides a framework for ethical decision-making.

Key Ethical Pillars

1. **Public Interest:** Software engineers shall act consistently with the public interest.
2. **Client and Employer:** Act in a manner that is in the best interests of the client and employer, consistent with the public interest.
3. **Product:** Ensure that products and related modifications meet the highest professional standards possible.
4. **Judgment:** Maintain integrity and independence in professional judgment.
5. **Management:** Managers shall promote an ethical approach to the management of software development.
6. **Profession:** Advance the integrity and reputation of the profession.
7. **Colleagues:** Be fair to and supportive of their colleagues.

8. **Self:** Participate in lifelong learning regarding the practice of their profession.

> ***Ethical Dilemma Example:*** *A software engineer is asked to bypass a security check to meet a tight deadline. According to the Code of Ethics, the engineer must prioritize the "Product" quality and "Public Interest" over the "Client/Employer"'s short-term deadline, as bypassing security could harm users.*
>
> ## *100 Difficult Multiple Choice Questions*

Part 1: Principles and Process Models (Questions 1-50)

1. Which principle of software engineering is most directly violated if a change in the user interface logic requires a complete rewrite of the database access layer? A. Generality B. Separation of Concerns C. Incremental Development D. Anticipation of Change

2. In the context of the SOLID principles, which principle is specifically aimed at preventing "fat" interfaces that force implementing classes to provide dummy implementations for methods they don't need? A. SRP B. OCP C. ISP D. DIP

3. A system is designed where high-level policy modules depend directly on low-level utility modules. Which design principle is being violated? A. Dependency Inversion Principle B. Liskov Substitution Principle C. Open/Closed Principle D. Modularity

4. Which software process model is characterized by a "risk-driven" approach, where each iteration begins with a risk analysis? A. Waterfall Model B. V-Model C. Spiral Model D. Scrum

5. In Scrum, who is primarily responsible for maintaining the Product Backlog and ensuring it is visible to all stakeholders? A. Scrum Master B. Development Team C. Project Manager D. Product Owner

6. Which of the following best describes the "V-Model" in software engineering? A. An extension of the waterfall model that links each development phase to a corresponding testing phase. B. A model that emphasizes rapid prototyping and user feedback. C. A circular model that focuses on continuous integration and deployment. D. A model used exclusively for hardware-software co-design.

7. The “Open/Closed Principle” suggests that a class should be: A. Open for modification but closed for extension. B. Open for extension but closed for modification. C. Open to all users but closed to external developers. D. Open for testing but closed for deployment.
8. In an Agile environment, what is the primary purpose of a “Sprint Retrospective”? A. To demonstrate the work completed during the sprint to stakeholders. B. To plan the tasks for the upcoming sprint. C. To inspect the team’s performance and identify improvements for the next sprint. D. To update the Product Backlog with new requirements.
9. Which architectural pattern is most closely associated with the principle of “Separation of Concerns” in web applications? A. Monolithic Architecture B. Model-View-Controller (MVC) C. Client-Server Architecture D. Peer-to-Peer Architecture
10. A software engineer is designing a system that must be highly reusable across different domains. Which principle should they prioritize? A. Consistency B. Generality C. Incremental Development D. Rigor
11. Which of the following is a key difference between Kanban and Scrum? A. Scrum uses time-boxed iterations (sprints), while Kanban uses a continuous flow model. B. Kanban has defined roles like Scrum Master, while Scrum does not. C. Scrum focuses on visual boards, while Kanban does not. D. Kanban requires a Product Owner, while Scrum does not.
12. The “Liskov Substitution Principle” is primarily concerned with: A. Interface design. B. Class inheritance and polymorphism. C. Database normalization. D. Network protocol compatibility.
13. Which DevOps practice involves automatically building and testing code changes as soon as they are committed to a shared repository? A. Continuous Deployment B. Continuous Integration C. Infrastructure as Code D. Microservices
14. In the Spiral Model, what happens if a high-risk factor cannot be mitigated during an iteration? A. The project proceeds to the next phase regardless. B. The project is immediately terminated. C. The iteration is repeated until the risk is gone. D. The requirements are changed to avoid the risk.

15. Which of the following is NOT one of the four core values of the Agile Manifesto? A. Individuals and interactions over processes and tools. B. Working software over comprehensive documentation. C. Following a plan over responding to change. D. Customer collaboration over contract negotiation.

16. What is the main disadvantage of the Waterfall model in modern software development? A. It is too complex to understand. B. It lacks a structured approach. C. It is difficult to accommodate changing requirements once the project has started. D. It requires too many developers.

17. “Software entities should be open for extension but closed for modification.” This statement defines which SOLID principle? A. SRP B. OCP C. LSP D. ISP

18. In Scrum, what is the “Definition of Done” (DoD)? A. A list of all features to be developed in the project. B. A shared understanding within the team of what it means for work to be complete. C. The date the project is expected to finish. D. The final sign-off from the customer.

19. Which model is most suitable for projects where requirements are well-defined and unlikely to change? A. Agile B. Waterfall C. Spiral D. XP

20. What does the “C” in CI/CD stand for? A. Component B. Continuous C. Collaborative D. Computed

21. Which principle emphasizes that a module should depend on abstractions rather than concrete implementations? A. Dependency Inversion B. Interface Segregation C. Separation of Concerns D. Modularity

22. In the context of software quality, what is “Maintainability”? A. The ability of the software to run on different platforms. B. The ease with which software can be modified to correct faults or improve performance. C. The degree to which the software protects information and data. D. The probability of failure-free operation for a specified time.

23. Which Agile methodology emphasizes “Pair Programming” as a core practice? A. Scrum B. Kanban C. Extreme Programming (XP) D. Lean Software Development

24. A “User Story” in Agile development typically follows which format? A. As a [role], I want [goal], so that [benefit]. B. The system shall [function]. C. If [condition], then [action]. D. [Feature Name]: [Description].

25. Which phase of the Waterfall model involves translating requirements into a technical blueprint for the software? A. Requirements Analysis B. System Design C. Implementation D. Integration

26. What is the primary focus of “DevSecOps”? A. Improving developer productivity. B. Integrating security practices into the DevOps pipeline. C. Reducing the cost of software licenses. D. Managing cloud infrastructure.

27. Which principle suggests that “a class should have only one reason to change”? A. Single Responsibility Principle B. Open/Closed Principle C. Liskov Substitution Principle D. Interface Segregation Principle

28. In a V-Model, which testing phase corresponds to the “Requirements Analysis” phase? A. Unit Testing B. Integration Testing C. System Testing D. Acceptance Testing

29. What is a “Burn-down Chart” used for in Scrum? A. To track the budget spent on the project. B. To visualize the amount of work remaining in a sprint. C. To list the bugs found during testing. D. To assign tasks to team members.

30. Which of the following is a characteristic of “Microservices Architecture”? A. All functions are contained within a single executable. B. Services are loosely coupled and independently deployable. C. It uses a single, shared database for all services. D. It is easier to test than monolithic architecture.

31. The term “Technical Debt” refers to: A. The money owed to software vendors. B. The cost of additional rework caused by choosing an easy solution now instead of a better approach that would take longer. C. The salaries of the development team. D. The cost of hardware upgrades.

32. Which principle is most effective at reducing the “ripple effect” when a change is made to one part of the system? A. Abstraction B. Modularity C. Consistency D. Generality

33. In XP, what is the purpose of “Test-Driven Development” (TDD)? A. To write tests after the code is finished. B. To write a failing test before writing the code to pass it. C. To automate all manual testing. D. To replace the need for professional testers.

34. Which of the following is a “State-of-the-Art” practice in software engineering as of 2026? A. Manual regression testing. B. Infrastructure as Code (IaC). C. Monthly

deployment cycles. D. Heavy upfront documentation.

35. What is the “Brooks’ Law”? A. Adding manpower to a late software project makes it later. B. Software quality is proportional to the number of testers. C. The cost of a bug increases exponentially over time. D. Requirements will always change.

36. Which principle promotes the idea that a subclass should be able to stand in for its parent class without causing errors? A. SRP B. OCP C. LSP D. ISP

37. In Scrum, who is responsible for removing “impediments” that hinder the team’s progress? A. Product Owner B. Scrum Master C. Development Team D. Stakeholders

38. Which model uses “Timeboxing” as a fundamental concept? A. Waterfall B. Agile C. Spiral D. Prototyping

39. What is the primary goal of “Refactoring”? A. To add new features to the software. B. To improve the internal structure of the code without changing its external behavior. C. To fix bugs in the software. D. To optimize the code for speed.

40. Which of the following best describes “Cohesion” in software design? A. The degree to which different modules depend on each other. B. The degree to which the elements inside a module belong together. C. The number of lines of code in a module. D. The speed at which a module executes.

41. High coupling between modules is generally considered: A. Good, because it means the modules work well together. B. Bad, because it makes the system harder to maintain and change. C. Irrelevant to software quality. D. Necessary for high-performance systems.

42. Which Agile framework uses “Work in Progress (WIP) limits”? A. Scrum B. Kanban C. XP D. Crystal

43. What is the “Mythical Man-Month”? A. A book about the history of programming. B. A concept that suggests human time and task time are perfectly interchangeable. C. A concept that highlights the complexities of managing software projects. D. A measure of developer productivity.

44. Which principle is violated if a class depends on a specific database driver rather than a database interface? A. Dependency Inversion Principle B. Interface

Segregation Principle C. Single Responsibility Principle D. Liskov Substitution Principle

45. In the context of software processes, what is “Evolutionary Prototyping”? A. Creating a prototype and then throwing it away. B. Developing a prototype that is gradually refined into the final system. C. Using AI to generate code prototypes. D. Testing a prototype with users before starting the real project.

46. Which of the following is a key benefit of “Continuous Deployment”? A. It eliminates the need for testing. B. It allows for faster feedback from real users. C. It reduces the need for version control. D. It guarantees that the software is bug-free.

47. “Don’t Repeat Yourself” (DRY) is a principle aimed at reducing: A. Code complexity. B. Redundancy in information and logic. C. The number of developers on a project. D. The time spent on meetings.

48. Which role in Scrum has no authority over the team but acts as a servant-leader? A. Product Owner B. Scrum Master C. Lead Developer D. Project Manager

49. What is the main focus of the “V-Model” during the “Integration Testing” phase? A. Testing individual components. B. Testing the interaction between integrated components. C. Testing the system against user requirements. D. Testing the system’s performance under load.

50. Which design principle is most closely related to the concept of “Information Hiding”? A. Abstraction B. Encapsulation C. Modularity D. Consistency

Part 2: Networking, Security, Tools, and Ethics (Questions 51-100)

51. Which layer of the OSI model is responsible for path determination and logical addressing (IP)? A. Data Link Layer B. Network Layer C. Transport Layer D. Session Layer

52. In a TCP/IP network, which protocol is used to map a known IP address to a physical MAC address? A. DNS B. DHCP C. ARP D. HTTP

53. Which of the following is a “connectionless” transport layer protocol? A. TCP B. UDP C. FTP D. SSH

54. What is the primary purpose of a “Subnet Mask” in networking? A. To identify the physical location of a computer. B. To distinguish the network portion of an IP address from the host portion. C. To encrypt data transmitted over the network. D. To speed up internet connections.
55. Which component of the CIA Triad is compromised if an unauthorized user gains access to a database of passwords? A. Confidentiality B. Integrity C. Availability D. Authenticity
56. A “Man-in-the-Middle” (MITM) attack primarily targets which aspect of security? A. Availability B. Confidentiality and Integrity C. Non-repudiation D. Physical Security
57. Which type of firewall inspects the actual content of the data packets rather than just the headers? A. Packet Filtering Firewall B. Stateful Inspection Firewall C. Application Level Gateway (Proxy Firewall) D. Circuit Level Gateway
58. What is the default port for the HTTPS protocol? A. 80 B. 21 C. 443 D. 22
59. In the context of computer security, what is “Social Engineering”? A. Writing code to automate social media posts. B. Manipulating individuals into divulging confidential information. C. Designing software for social networks. D. Analyzing the social impact of technology.
60. Which encryption method uses the same key for both encryption and decryption? A. Asymmetric Encryption B. Symmetric Encryption C. Hashing D. Digital Signatures
61. What is the primary function of a “Router” in a network? A. To connect multiple devices within a single LAN. B. To forward data packets between different networks. C. To provide power to network devices. D. To store website data.
62. Which of the following is a common threat where an attacker injects malicious scripts into a trusted website? A. SQL Injection B. Cross-Site Scripting (XSS) C. Denial of Service (DoS) D. Buffer Overflow
63. The “Principle of Least Privilege” suggests that: A. All users should have administrative access. B. Users should be given only the minimum levels of access necessary to perform their job functions. C. Security should be the responsibility of the lowest-paid employees. D. Software should be as cheap as possible.

64. Which network topology connects all devices to a central hub or switch? A. Bus Topology B. Ring Topology C. Star Topology D. Mesh Topology
65. What does the "A" in the CIA Triad stand for? A. Authentication B. Availability C. Authorization D. Accountability
66. Which tool is essential for managing different versions of source code and allowing multiple developers to work on the same project? A. Jenkins B. Git C. Docker D. Jira
67. What is the purpose of an "Integrated Development Environment" (IDE)? A. To host websites. B. To provide a centralized set of tools for software development. C. To manage network traffic. D. To backup data.
68. Which of the following is a "Build Automation" tool? A. Maven B. VS Code C. Selenium D. Trello
69. In Git, which command is used to combine changes from one branch into another? A. git commit B. git push C. git merge D. git checkout
70. What is the primary use of "Docker" in modern software engineering? A. To write code in multiple languages. B. To package applications and their dependencies into containers for consistent deployment. C. To manage project timelines. D. To design user interfaces.
71. Which testing tool is commonly used for automating web browser interactions? A. JUnit B. Selenium C. Postman D. Wireshark
72. What is "Static Code Analysis"? A. Testing the code while it is running. B. Analyzing the source code without executing it to find potential errors or style issues. C. Measuring the speed of the code. D. Checking the code's impact on network traffic.
73. Which of the following is an ethical responsibility of a software engineer according to the ACM Code of Ethics? A. To always follow the employer's instructions, even if they are unethical. B. To prioritize the public interest in all professional decisions. C. To maximize profit for the company at all costs. D. To avoid learning new technologies.
74. "Intellectual Property" (IP) in software engineering refers to: A. The physical computers used by developers. B. Legal rights resulting from intellectual activity

in the industrial, scientific, literary, and artistic fields. C. The office space where the software is developed. D. The internet bandwidth used by the team.

75. If a software engineer discovers a critical security flaw in a product they are developing, what should they do first? A. Ignore it to meet the deadline. B. Report it to their manager or the appropriate authority within the organization. C. Post about it on social media. D. Sell the information to a third party.

76. Which of the following is an example of an “Ethical Dilemma” in software engineering? A. Choosing between Java and Python for a project. B. Deciding whether to report a bug that might delay a product launch. C. Selecting a color scheme for a website. D. Deciding which office chair to buy.

77. The concept of “Informed Consent” in data privacy means: A. Users are forced to share their data. B. Users are given clear information about how their data will be used and must agree to it. C. Data is shared without the user’s knowledge. D. Only the government can access user data.

78. Which of the following is NOT a professional norm in software engineering? A. Writing clean and documented code. B. Respecting the confidentiality of clients. C. Plagiarizing code from other developers. D. Continuous professional development.

79. What is “Whistleblowing” in a professional context? A. Reporting a colleague for being late. B. Disclosing an employer’s unethical or illegal activities to an outside authority. C. Playing a musical instrument at work. D. Complaining about the office coffee.

80. According to the IEEE Code of Ethics, software engineers should: A. Accept bribes if they are small. B. Avoid conflicts of interest. C. Discourage colleagues from learning new skills. D. Hide their mistakes from clients.

81. Which layer of the OSI model handles data compression and encryption? A. Application B. Presentation C. Session D. Transport

82. A “Denial of Service” (DoS) attack primarily affects which part of the CIA triad? A. Confidentiality B. Integrity C. Availability D. Accountability

83. What is the purpose of a “Digital Certificate”? A. To prove a user’s identity to a website. B. To verify the ownership of a public key. C. To encrypt an entire hard drive. D. To speed up network downloads.

84. Which protocol is used for sending emails? A. POP3 B. IMAP C. SMTP D. HTTP
85. In a Local Area Network (LAN), what is the typical maximum distance for a standard Ethernet cable (Cat5e/Cat6) without a repeater? A. 10 meters B. 100 meters C. 1000 meters D. 50 meters
86. Which of the following is a “system software”? A. Microsoft Word B. Google Chrome C. Linux Kernel D. Adobe Photoshop
87. What does “SQL Injection” involve? A. Sending too many emails to a server. B. Inserting malicious SQL statements into an entry field for execution. C. Stealing a physical hard drive. D. Cracking a Wi-Fi password.
88. “Two-Factor Authentication” (2FA) improves security by: A. Requiring two different passwords. B. Requiring something the user knows and something the user has. C. Requiring two different users to log in. D. Making the password twice as long.
89. Which Git command is used to upload local repository content to a remote repository? A. git pull B. git fetch C. git push D. git add
90. What is the main purpose of “Unit Testing”? A. To test the entire system as a whole. B. To test individual components or functions in isolation. C. To test the software’s performance on different devices. D. To test the user interface design.
91. Which of the following is a project management tool commonly used in Agile teams? A. Wireshark B. Jira C. GDB D. Postman
92. “Refactoring” should be done: A. Only when a bug is found. B. Regularly to maintain code quality. C. Only by senior developers. D. At the very end of the project.
93. What is “Pair Programming”? A. Two developers writing two different programs. B. Two developers working together at one workstation. C. Two developers competing to finish a task. D. One developer writing code and another writing documentation.
94. Which of the following is a “non-functional requirement”? A. The system shall allow users to log in. B. The system shall process 1000 transactions per second. C. The system shall calculate the total price. D. The system shall send a confirmation email.

95. What is the purpose of a “Code Review”? A. To punish developers for making mistakes. B. To improve code quality and share knowledge among team members. C. To increase the lines of code in the project. D. To replace the need for automated testing.
96. “Software as a Service” (SaaS) refers to: A. Selling software on physical discs. B. Delivering applications over the internet as a service. C. Hiring developers to write custom software. D. Providing hardware for software development.
97. Which of the following is an example of “Open Source” software? A. Microsoft Windows B. Adobe Reader C. Mozilla Firefox D. Apple iOS
98. The “Public” pillar of the ACM Code of Ethics states that software engineers shall: A. Only care about the public if it pays them. B. Disclose any potential danger to the user, the public, or the environment. C. Hide all information from the public. D. Avoid working on public projects.
99. “Competence” in professional ethics means: A. Being better than all your colleagues. B. Only undertaking tasks that you are qualified to perform. C. Pretending to know everything. D. Working as fast as possible.
100. What is the final step in most software development life cycles? A. Coding B. Testing C. Maintenance and Evolution D. Requirements Gathering

Answer Key for 100 Multiple Choice Questions

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	B	26	B	51	B	76	B
2	C	27	A	52	C	77	B
3	A	28	D	53	B	78	C
4	C	29	B	54	B	79	B
5	D	30	B	55	A	80	B
6	A	31	B	56	B	81	B
7	B	32	B	57	C	82	C

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
8	C	33	B	58	C	83	B
9	B	34	B	59	B	84	C
10	B	35	A	60	B	85	B
11	A	36	C	61	B	86	C
12	B	37	B	62	B	87	B
13	B	38	B	63	B	88	B
14	B	39	B	64	C	89	C
15	C	40	B	65	B	90	B
16	C	41	B	66	B	91	B
17	B	42	B	67	B	92	B
18	B	43	C	68	A	93	B
19	B	44	A	69	C	94	B
20	B	45	B	70	B	95	B
21	A	46	B	71	B	96	B
22	B	47	B	72	B	97	C
23	C	48	B	73	B	98	B
24	A	49	B	74	B	99	B
25	B	50	B	75	B	100	C